



Analytics in the Era of AI

Deniz Atma
Cloud Solution Architect





Deniz Atma

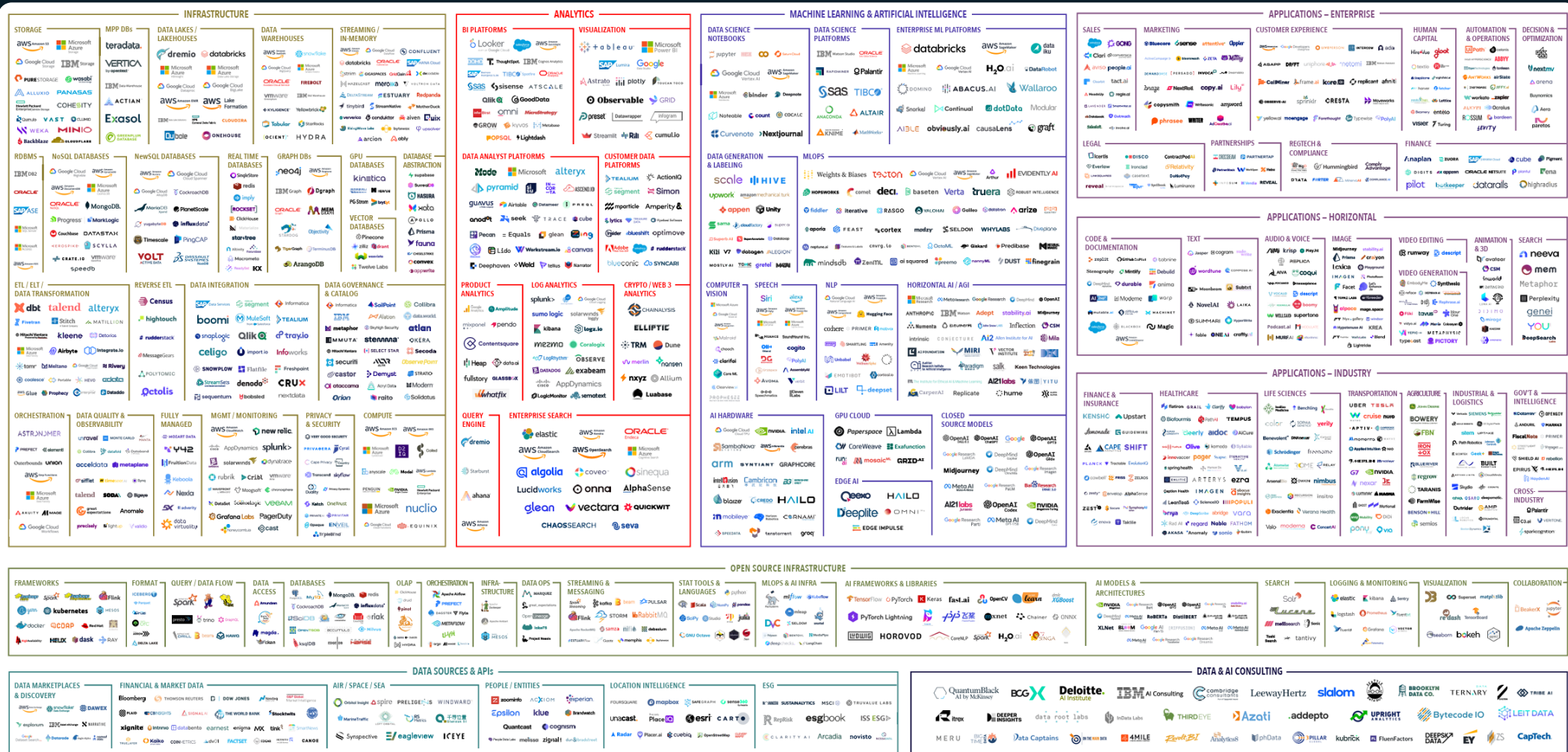
Cloud Solution Architect

Combining > 10 years of IT experience – especially in the field of Business Intelligence solutions and drove success for several enterprise companies. Trusted advisor for Azure-based solutions and Microsoft technologies.

- Azure architectures
- Analytics
- SQL / NoSQL databases
- Microsoft Fabric
- Power Platform
- Data engineering

Data is the **fuel** that powers AI

But face immense complexity



Announcing



Microsoft Fabric

Data analytics for the era of AI



Microsoft Fabric

Data analytics for the era of AI

Complete
Analytics
Platform

Lake Centric
and Open

Empower
Every
Business User

AI
Powered



Microsoft Fabric: Big Data Analytics Platform

Eliminate Data Siloes and duplications



Data Governance: Microsoft Purview Evolution

Compute Engines: SaaS-ified, AI-powered, under same business model



Data
Integration



Data
Warehouse



Data
Engineering



Data
Science



Real Time
Intelligence



Business
Intelligence



Applied
Observability



OneLake

One copy of all your data; Open formats (Parquet, Delta); One Security; Leverage existing data multi-cloud

Enterprise Data Sources – Multi-Cloud | On-premises | External



Unstructured



Structured



Semi-structured



Azure



AWS

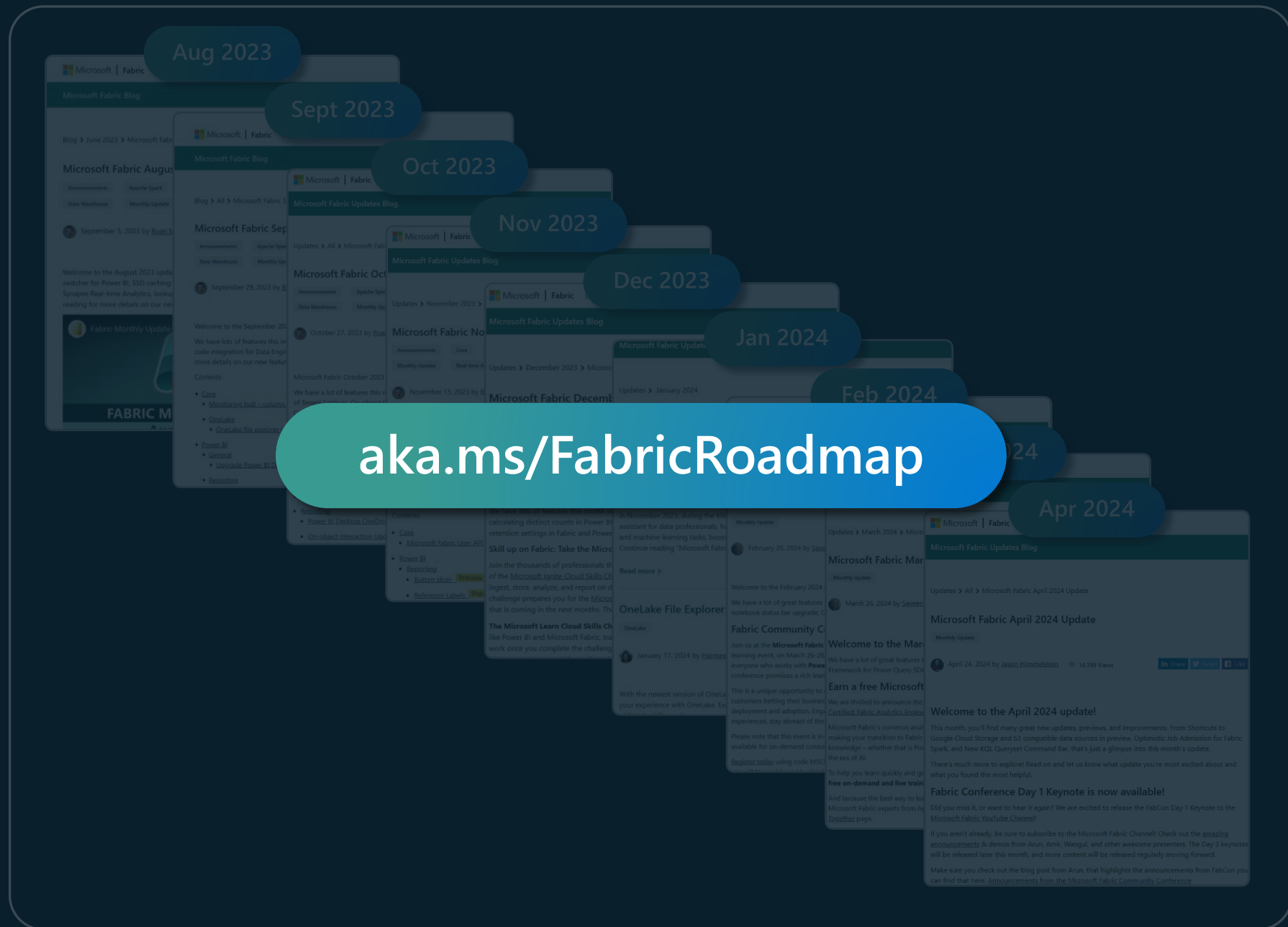


Google

Rapid pace
of innovation:

Weekly Fabric Releases

aka.ms/FabricRoadmap





Microsoft Fabric

The unified data platform for the era of AI



Data
Factory



Data
Engineering



Data
Warehouse



Data
Science



Real-Time
Intelligence



Power BI



AI



OneLake



Purview



Microsoft Fabric

The unified data platform for the era of AI



Data
Factory



Data
Engineering



Data
Warehouse



Data
Science



Real-Time
Intelligence



Power BI



Partner
workloads



AI



OneLake



Purview



Microsoft Fabric

The unified data platform for the era of AI



Data Factory



Power BI



Data Engineering



Real-Time Intelligence



Data Warehouse



Data Science



COGNITE



esri



Informatica



LSEG

neo4j



NVIDIA

Profisee

quantexa

sas

striim

teradata.

and more...



AI



OneLake



Purview

Public preview

Microsoft Fabric Workload Development kit

Acquire new customers with your own Fabric workload

Seamless user
experience

OneLake
integration

Enhanced customer
discoverability

Marketplace-based
monetization



Microsoft Fabric

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Familiar and intuitive

Built into Microsoft 365

Insight to action

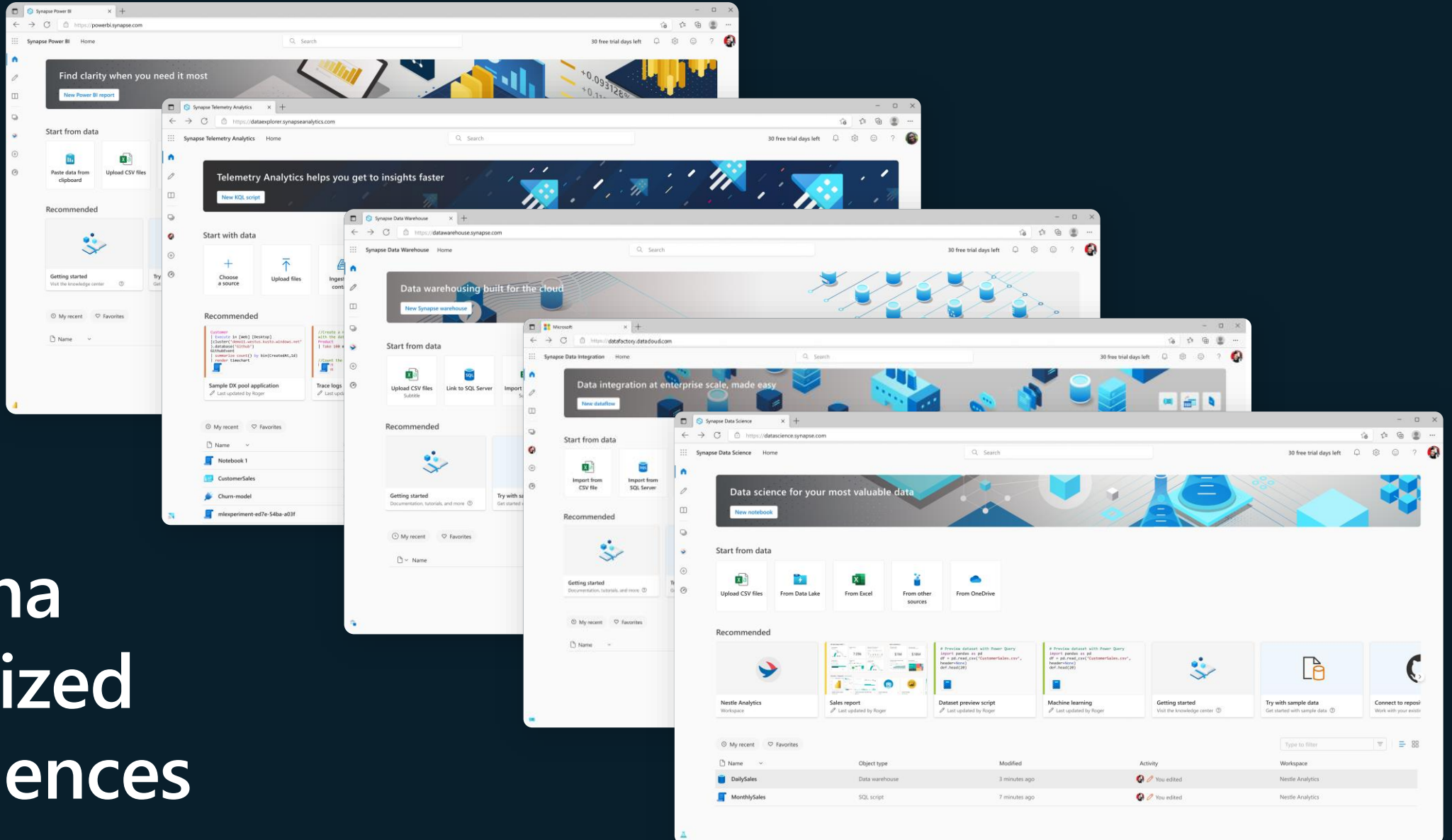
AI Powered

Copilot accelerated

GPT on your data

AI-driven insights

Persona optimized experiences



Demo

Persona Optimized Experiences

< >

You favorited this



Azure Data LT Dashboar

Filter

 Name	Type	Opened	Location	Endorsement	Sensitivity
 Contoso Outdoors Sales 	Workspace	39 seconds ago	Workspaces	—	—
 Customer feedback finder 	Report	55 seconds ago	PBI CAT Feedback	—	General ⓘ
 Workforce Planning FY23	Report	a minute ago	Workforce Planning	—	Confidential\Microsoft Extended ⓘ
 My workspace 	Workspace	a minute ago	Workspaces	—	—
 User Source Reporting	Report	a minute ago	—	—	Highly Confidential\Microsoft Ex... ⓘ
 ESG Indicators 	Report	a minute ago	Gartner Data and Analytics Su...	—	General ⓘ
 Hotel Customer Retention 	Report	a minute ago	My workspace	—	Confidential\Microsoft FTE ⓘ
 Workforce Demographics	Report	20 minutes ago	Workforce Planning	—	General ⓘ

SaaS

"It just works"

5x5

Frictionless onboarding

Instant Provisioning

Quick results w/ Intuitive UX

Success by Default

Minimal knobs

Auto optimized

Auto Integrated

Centralized Administration

Tenant-wide governance

Centralized
security management

Compliance built-in

Built-in CI/CD

Scales to the most demanding projects

The screenshot displays the Microsoft Fabric Power BI interface for a workspace named 'Contoso Sales Analytics - Dev'. The interface includes a left-hand navigation pane with icons for Home, Create, Browse, OneLake data hub, Apps, Metrics, Monitoring hub, Deployment pipelines, Learn, and Workspaces. The main area shows a table of assets with columns for Name, Git status, Type, Task, Owner, Refreshed, Next refresh, Endorsement, Sensitivity, and Include. The 'Daily Sales Update' data pipeline is highlighted, showing it is 'Synced' and owned by Patrick LeBlanc. Other assets include 'ContosoAnalytics' (Lakehouse, Semantic model, SQL analytics end...), 'Customer Listing' (Paginated Report), 'Employee Information' (Report, Semantic model), 'Employee Sales' (Report, Semantic model), 'Fraud Analysis and Detection' (Paginated Report), 'Initialize the Gold Zone' (Notebook), 'Load Bronze Zone' (Notebook), 'Load Silver Zone' (Notebook), 'Store Sales Analytics' (Report, Semantic model), and 'Store Sales Analytics' (Semantic model).

Name	Git status	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Include
ContosoAnalytics	Synced	Lakehouse		Patrick LeBlanc	—	—	—	Confidential\Micro...	
ContosoAnalytics	—	Semantic model (...)		Contoso Sales An...	3/13/24, 10:25:32 AM	N/A	—	Confidential\Micro...	
ContosoAnalytics	—	SQL analytics end...		Contoso Sales An...	3/18/24, 10:20:01 AM	N/A	—	Confidential\Micro...	
Customer Listing	Synced	Paginated Report		Contoso Sales An...	—	—	—	Confidential\Micro...	
Daily Sales Update	Synced	Data pipeline		Patrick LeBlanc	—	—	—	Confidential\Micro...	
Employee Information	Synced	Report		Contoso Sales An...	3/18/24, 10:03:36 AM	—	—	Non-Business	
Employee Information	Synced	Semantic model		Contoso Sales An...	3/18/24, 10:03:36 AM	N/A	—	Non-Business	
Employee Sales	Synced	Report		Contoso Sales An...	3/18/24, 10:03:24 AM	—	—	Non-Business	
Employee Sales	Synced	Semantic model		Contoso Sales An...	3/18/24, 10:03:24 AM	N/A	—	Non-Business	
Fraud Analysis and Detection	Synced	Paginated Report		Contoso Sales An...	—	—	—	Confidential\Micro...	
Initialize the Gold Zone	Synced	Notebook		Patrick LeBlanc	—	—	—	Confidential\Micro...	
Load Bronze Zone	Synced	Notebook		Patrick LeBlanc	—	—	—	Confidential\Micro...	
Load Silver Zone	Synced	Notebook		Patrick LeBlanc	—	—	—	Confidential\Micro...	
Store Sales Analytics	Synced	Report		Contoso Sales An...	3/13/24, 12:11:38 PM	—	—	Non-Business	
Store Sales Analytics	Synced	Semantic model		Contoso Sales An...	3/13/24, 12:11:38 PM	N/A	—	Non-Business	



Microsoft Fabric

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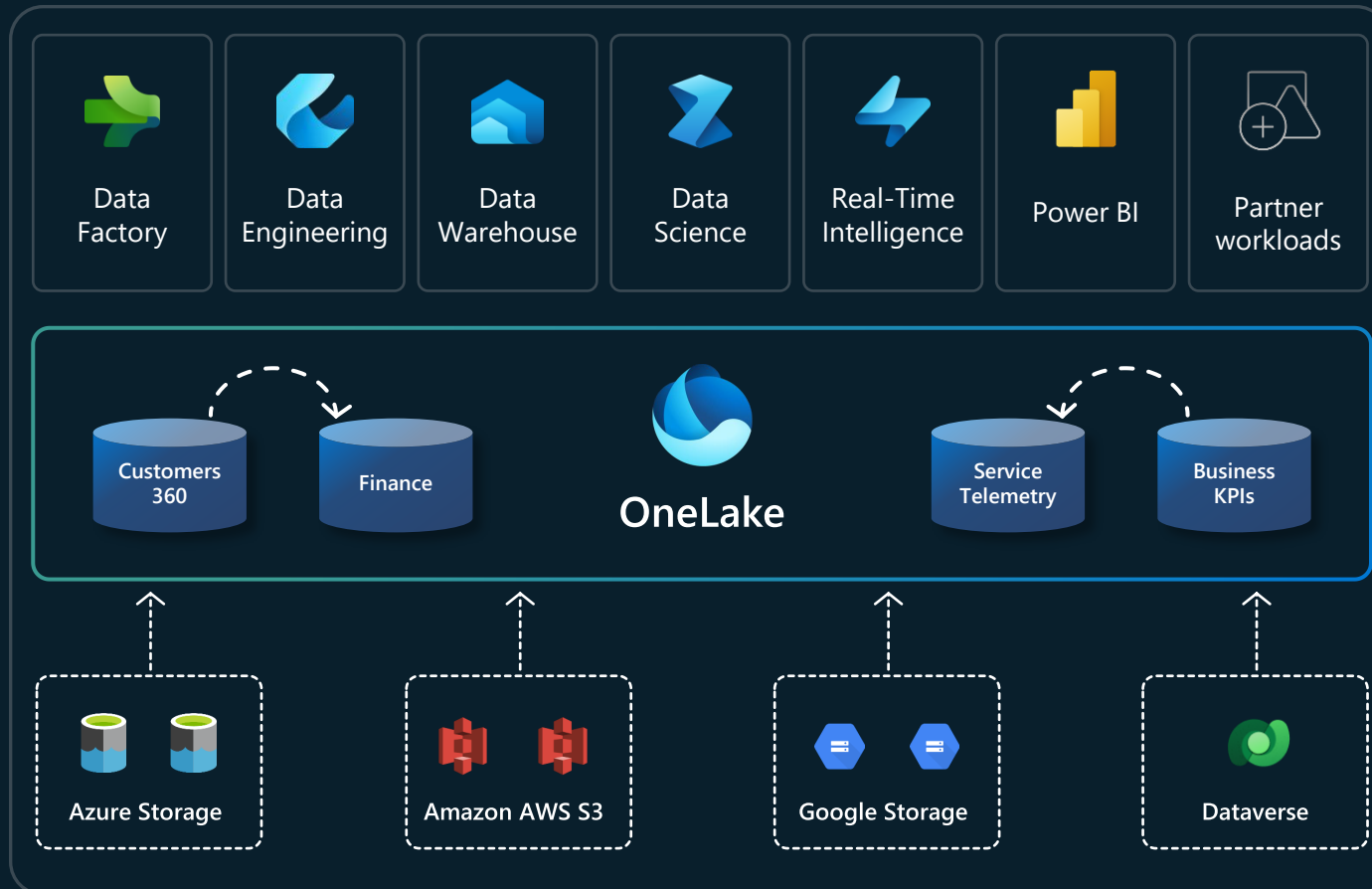
Copilot accelerated

GPT on your data

AI-driven insights

Unifying data in OneLake

Multi-Cloud Shortcuts



Sharing data in OneLake is as easy as sharing files in OneDrive, removing the needs for data duplication

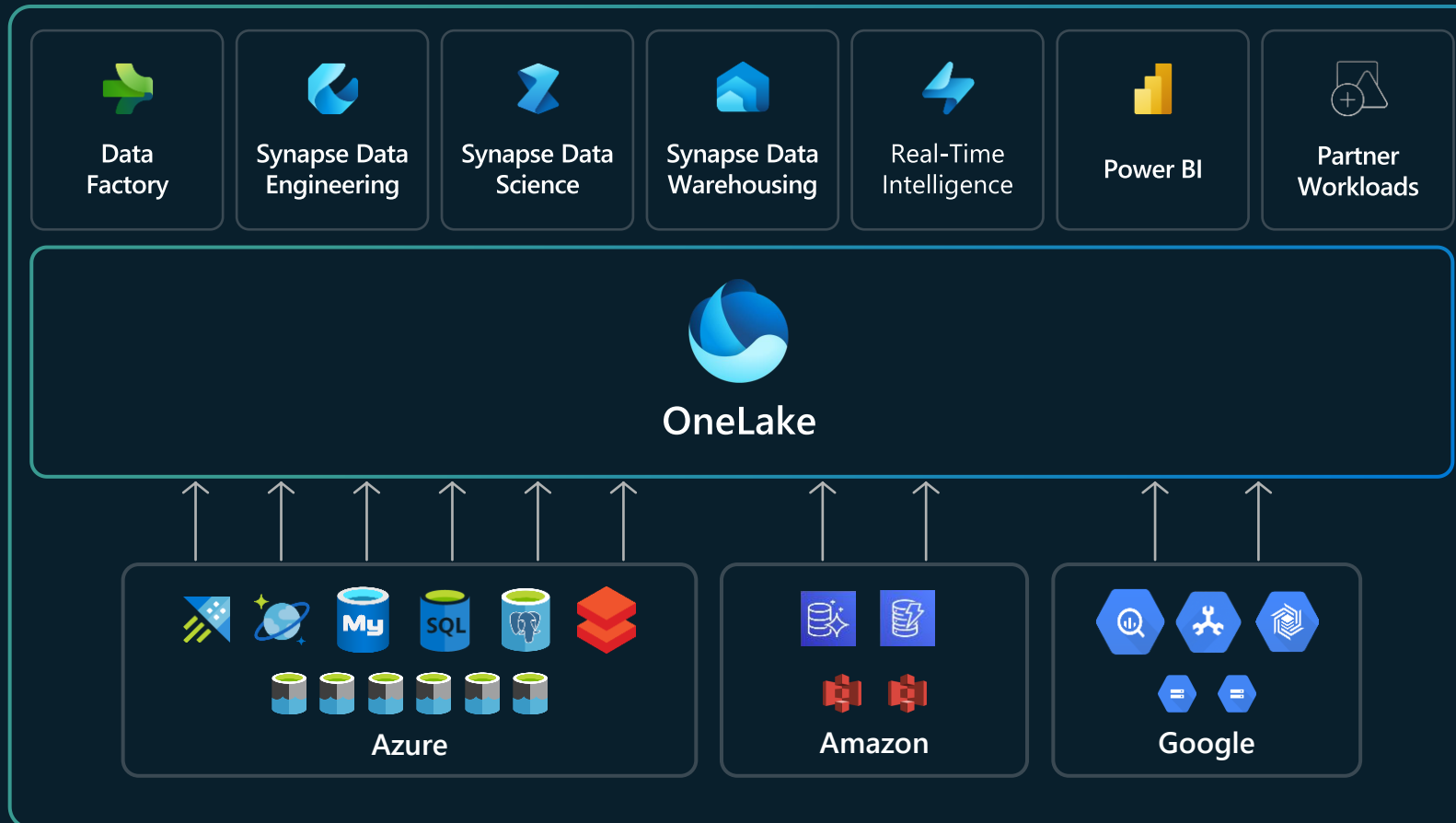
With **shortcuts**, data throughout OneLake can be composed together without any data movement

Shortcuts also allow instant linking of data already existing in Azure and in other clouds, without any data duplication and movement, making **OneLake** the first **multi-cloud data lake**

With support for industry standard APIs, OneLake data can be directly accessed by any application or service

Unifying data in OneLake

Mirroring of External Databases



Factionless linking of external databases, with full replicas created with a couple of clicks

Available for both multi-cloud and on-premises databases

Real time updates of the replicas using the CDC feeds of the database

Data is stored in Delta Parquet tables, with all Fabric services instantly available

Free Mirroring Storage for Replicas



OneLake

Demo

Shortcuts, Mirroring



OneLake



Delta Lake
Format



OneLake



Delta Lake
Format

Announcing



Iceberg
Format

Transparent simultaneous support for Delta
Lake and Iceberg formats

Do more with your data with open standards

Unify your data in OneLake

Bi-directional data access

Seamless access from M365
and Copilot



Microsoft Fabric



Salesforce

Do more with your data with open standards

Iceberg support in OneLake

Bi-directional data access

Seamless access from M365
and Copilot

Expanded Partnership



Microsoft Fabric



Snowflake

Demo



Microsoft Fabric



Snowflake



OneLake

Synapse Data Engineering

https://msit.powerbi.com/groups/05c5c8b8-3435-4caa-b542-092cd21c94f9/list?experience=data-engineering

Synapse Data Engineering

Contoso Sales Excellence

Search

Trial: 24 days left

Home

Create

Browse

OneLake data hub

Monitor

Real-Time hub

Workspaces

Contoso Sales...

Data Engineering

Contoso Sales Excellence

Create deployment pipeline

Update app

Manage access

+ New

Upload

Filter by keyword

Filter (1)

Clear all

Lakehouse

Name	Type	Task	Owner	Refreshed
ContosoOutdoors_CustomerSupport	Lakehouse	—	Patrick Baumgartn...	—
ContosoOutdoors_CustomerSupport	Semantic model (...)	—	Contoso Sales Exc...	10/18/23, 2:11:33 PM
ContosoOutdoors_CustomerSupport	SQL analytics end...	—	Contoso Sales Exc...	10/27/23, 12:55:04 PM
ContosoSalesKPILakehouse	Lakehouse	—	Patrick Baumgartn...	—
ContosoSalesKPILakehouse	Semantic model (...)	—	Contoso Sales Exc...	7/17/23, 11:41:40 AM
ContosoSalesKPILakehouse	SQL analytics end...	—	Contoso Sales Exc...	—
CustomerReviews	Lakehouse	—	Patrick Baumgartn...	—
CustomerReviews	Semantic model (...)	—	Contoso Sales Exc...	7/6/23, 10:33:09 PM
CustomerReviews	SQL analytics end...	—	Contoso Sales Exc...	—
OrderStatus	Lakehouse	—	Patrick Baumgartn...	—

Databases

https://app.snowflake.com/qmbdepq/vr30238/#/data/databases

snowflake

Create

Projects

Data

Databases

Add Data

Data Products

AI & ML

Monitoring

Admin

Databases

+ Database

5 Databases

Search

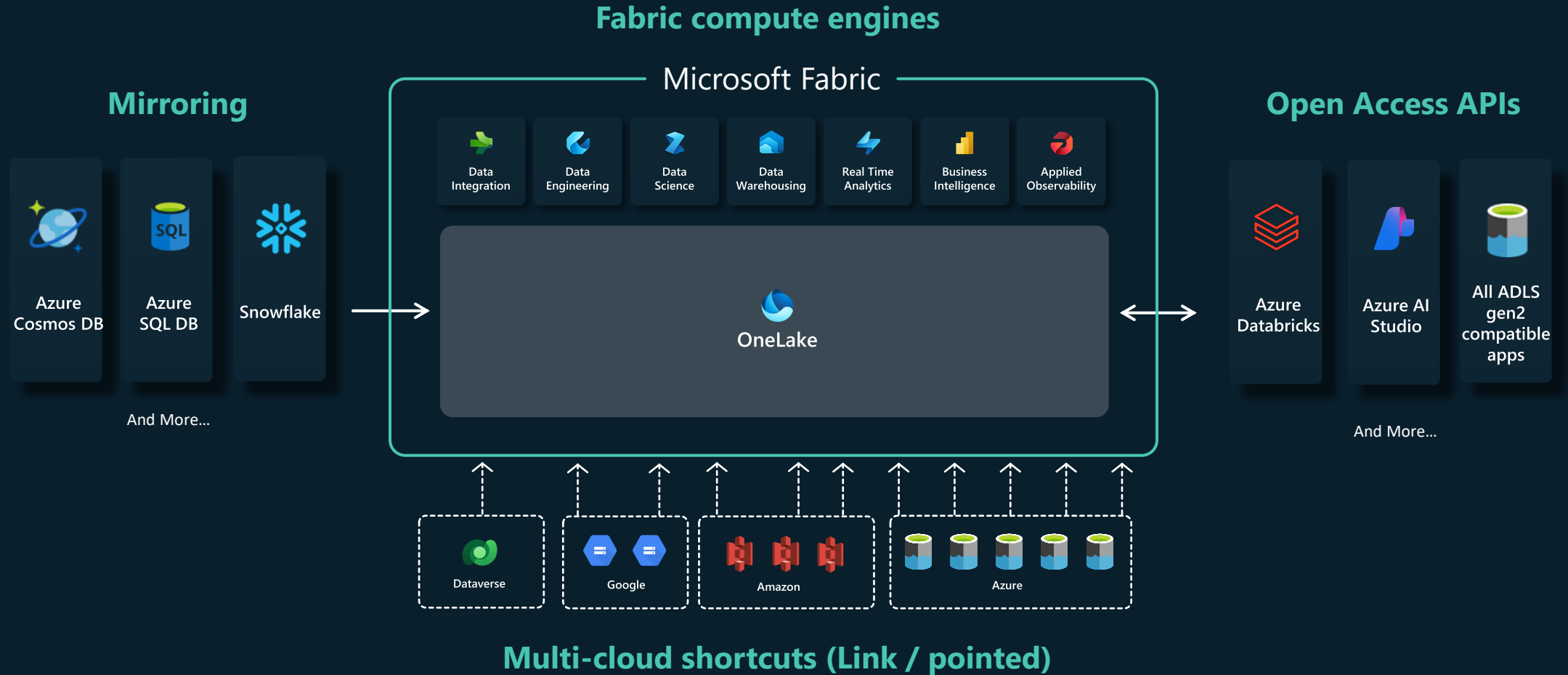
Source All

Refresh

NAME	SOURCE	OWNER	CREATED
CONTOSO_DATABASE	Local	ACCOUNTADMIN	1 day ago
CONTOSO_LOYALTY_PROGRAM	Local	ACCOUNTADMIN	1 day ago
CONTOSO_OUTDOORS_MARKETING	Local	ACCOUNTADMIN	1 day ago
CONTOSO_OUTDOORS_MEMBERS	Local	ACCOUNTADMIN	1 day ago
SNOWFLAKE	Share	—	1 week a...

Single View & Multi Cloud Connected Data Sources

Why is Microsoft Fabric a game changer





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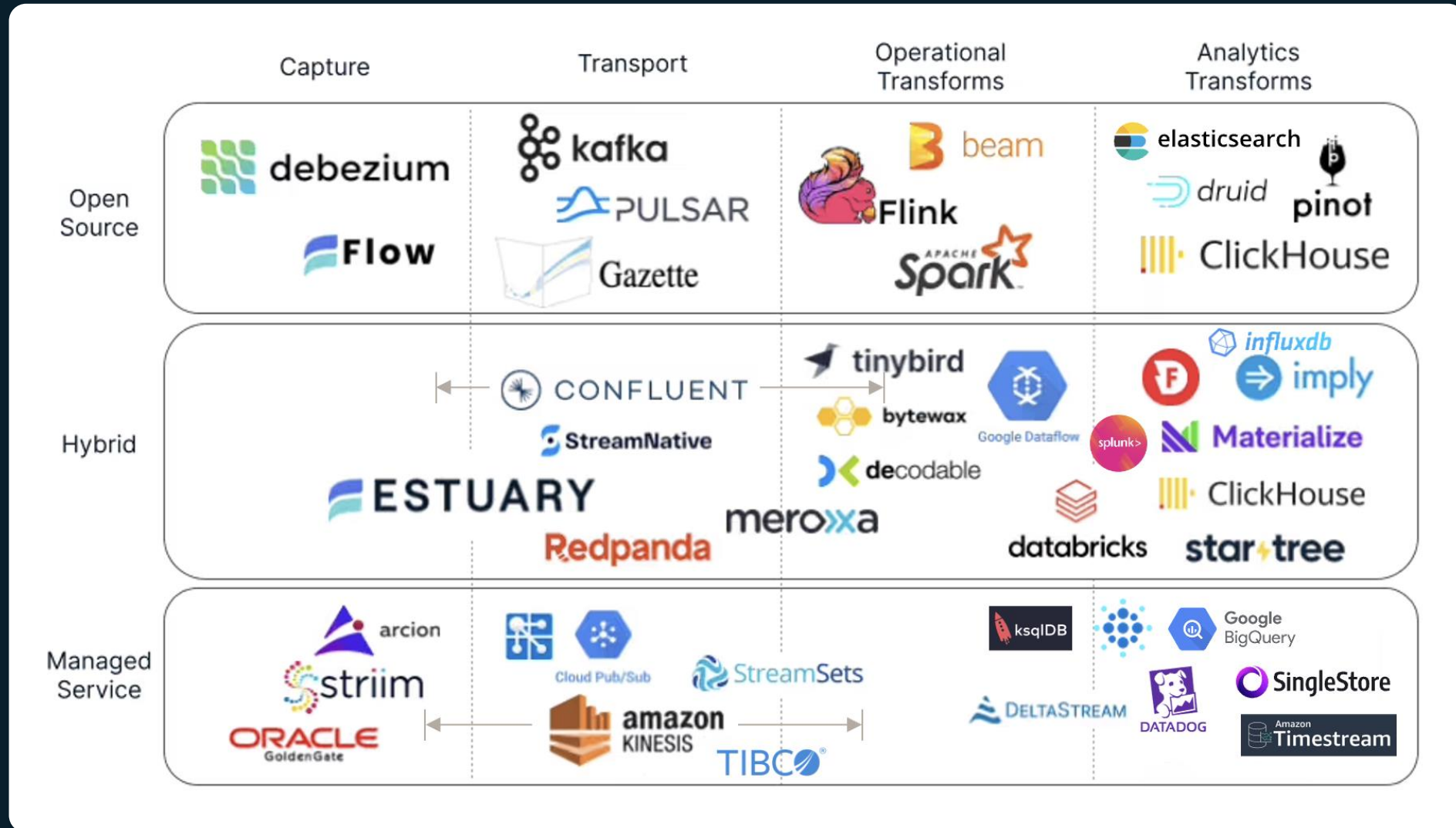
AI Powered

Copilot accelerated

GPT on your data

AI-driven insights

Fragmented landscape for real-time data



Announcing



Real-Time Intelligence

at Microsoft Build 2024



Microsoft Fabric

The unified data platform for the era of AI



Data
Factory



Data
Engineering



Data
Warehouse



Data
Science



Real-Time
Intelligence



Power BI



Partner
workloads



AI

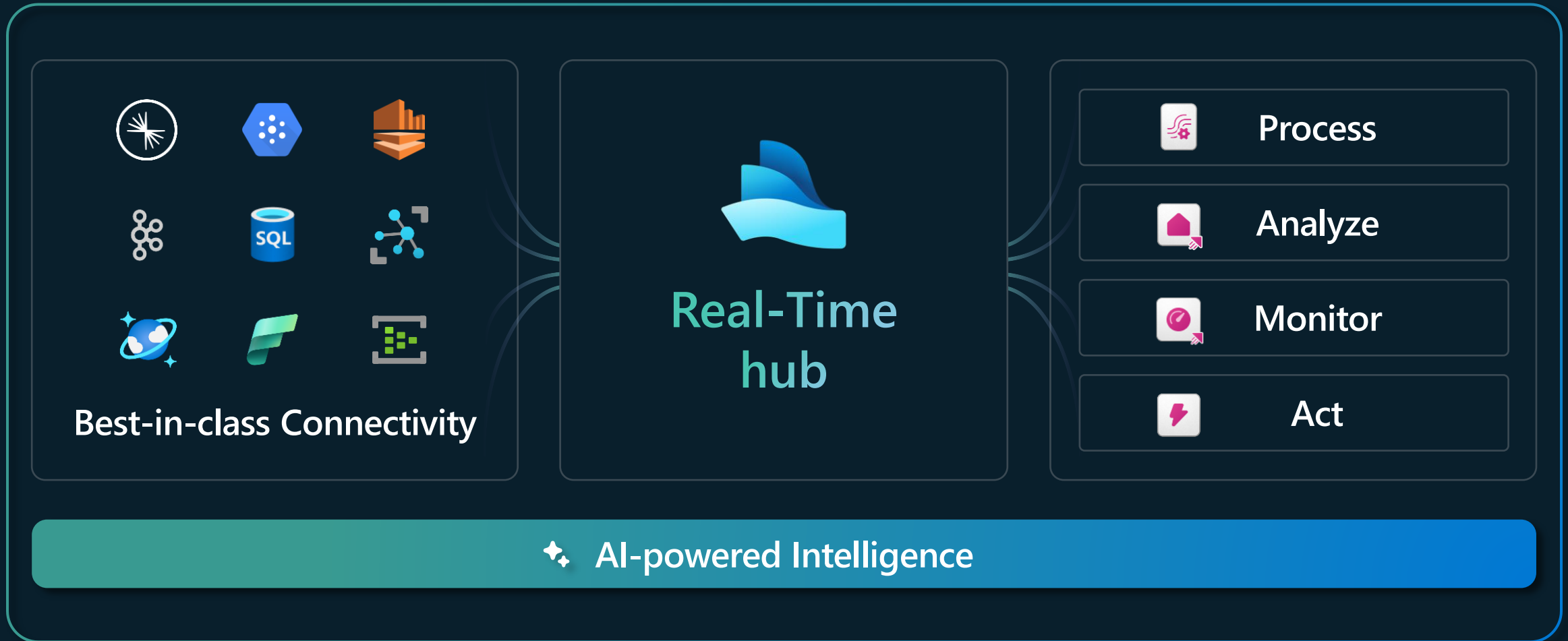


OneLake



Purview

Real-Time Intelligence in Microsoft Fabric



Home

Create

Browse

OneLake data hub

Monitor

Real-Time hub

Workspaces

Contoso Operations

Contoso Operations

+ New

Upload

Add (preview)

Streaming data in...

Get data

1 item

+ New item

Data track

Track data

2 items

+ New item

Store streaming da...

Store data

2 items

+ New item

Near real-time dat...

Visualize data

1 item

+ New item

Event analytics

Task flow details

Seamlessly integrate both real-time and time-based data into a unified system. Ingest data from diverse sources to extract valuable insights and promptly respond to changing conditions. This empowers data-driven decisions based on the most up-to-date information available.

Tasks

4 Tasks

Streaming data ingest

Store streaming data

Data track

Near real-time data visual...

Name	Type	Task	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
Routes	Eventstream	Streaming data...	Tessa Kloster (PAL...	—	—	—	Confidential\Micro...	
Operations	Eventhouse	Store streamin...	Tessa Kloster (PAL...	—	—	—	Confidential\Micro...	
Operations	KQL Database	Store streamin...	—	—	—	—	Confidential\Micro...	
Deliveries	Real-Time Dashbo...	Near real-time ...	Tessa Kloster (PAL...	—	—	—	Confidential\Micro...	
ExploreDeliveries	KQL Queryset	Data track	Tessa Kloster (PAL...	—	—	—	Confidential\Micro...	
ExploreDeliveriesQS	KQL Queryset	Data track	Tessa Kloster (PAL...	—	—	—	Confidential\Micro...	

Demo

Office Integration

MSMechanics_Fabric 

MS Mechanics Demo WS

+ New ▾

Upload

⚙ Workspace settings

🔍 Filter by keyword

Filter



	Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity
	aisample 	Experiment	Marcos Campos	—	—	—	Confidential\Micro..
	Customer churn-892 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	Dataflow 1	Dataflow Gen2	Abhishek Narain	—	N/A	—	—
	Export Tables 	Notebook	Markus Cozowicz	—	—	—	Confidential\Micro..
	Ingest data into bronze lh 	Data pipeline	Abhishek Narain	—	—	—	Confidential\Micro..
	NB_Copilot 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	NB_Copilot_Mechanics 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	NB_Copilot_Mechanics 	Experiment	Nellie Gustafsson	—	—	—	Confidential\Micro..
	NB_Copilot_Raj 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	NG_Copilot_Test 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	NG_Copilot_Test Copy 	Notebook	Marcos Campos	—	—	—	Confidential\Micro..
	Notebook 1 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..
	Notebook 2 	Notebook	Marcos Campos	—	—	—	Confidential\Micro..
	Notebook 3 	Notebook	Nellie Gustafsson	—	—	—	Confidential\Micro..



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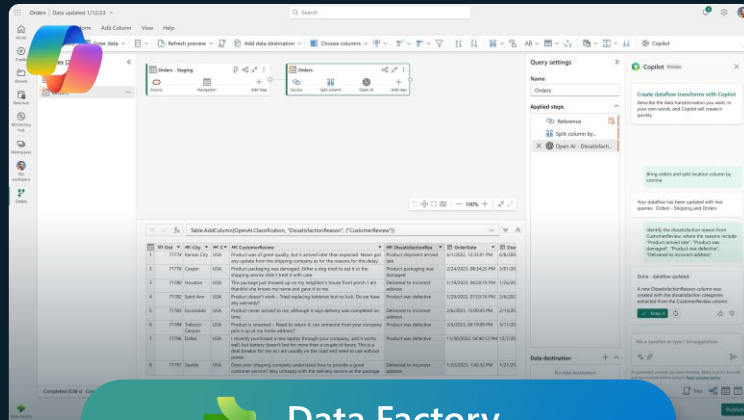
AI Powered

Copilot accelerated

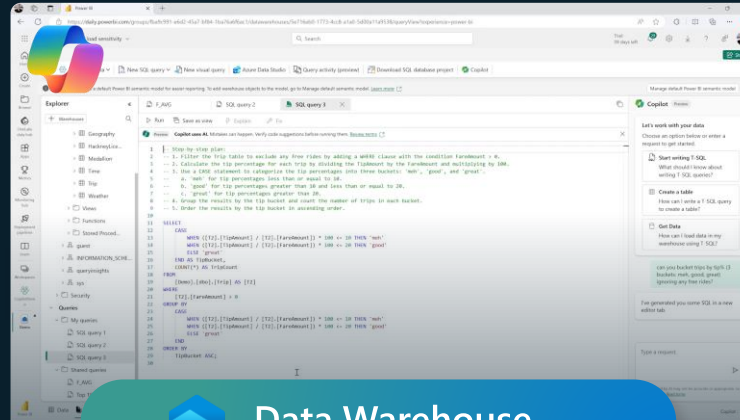
GPT on your data

AI-driven insights

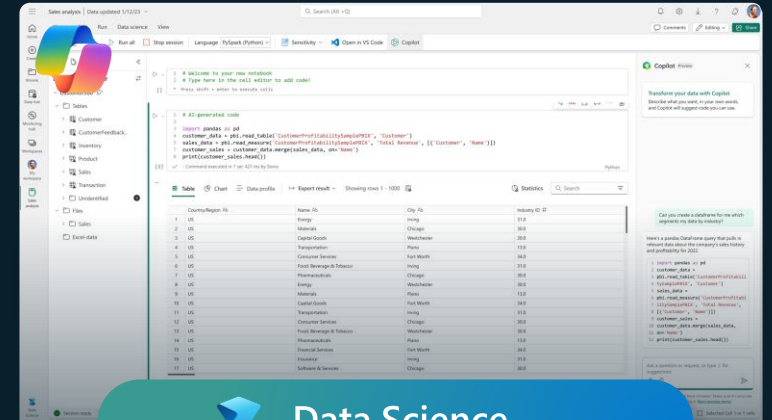
Copilot Integrated in every Microsoft Fabric Experience



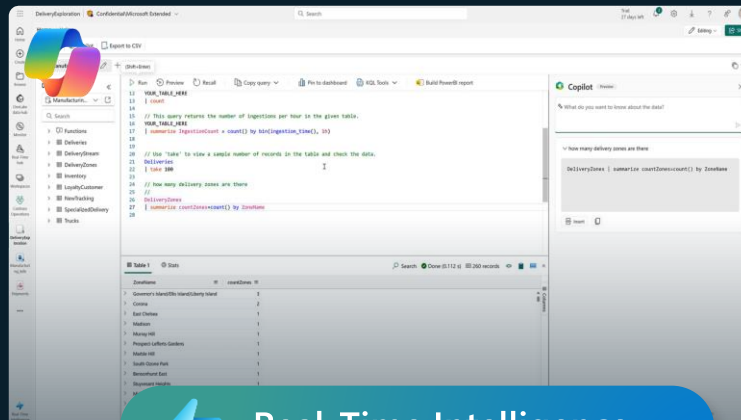
Data Factory



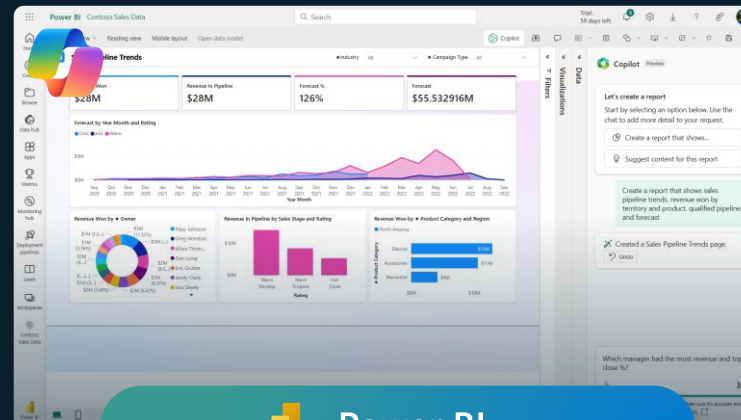
Data Warehouse



Data Science



Real-Time Intelligence



Power BI

Demo

Copilot in Data Science

Inspect

Demo

Copilot in Power BI

Contoso Outdoors Bike Rentals

+ New

Upload

Create deployment pipeline

Create app

Manage access

Filter by keyword

Filter

	Name	Type	Owner	Refreshed	Next refresh	Endorsement	Sensitivity	Included in app
	ContosoOutdoorsBikeRental	Lakehouse	Patrick Baumgartn...	—	—	—	Confidential\Micro... ⓘ	
	ContosoOutdoorsBikeRental	Semantic model (...)	Contoso Outdoor...	11/6/23, 5:04:23 PM	N/A	—	Confidential\Micro... ⓘ	
	ContosoOutdoorsBikeRental	SQL analytics end...	Contoso Outdoor...	—	N/A	—	Confidential\Micro... ⓘ	
	E-Bike Data Import	Dataflow Gen2	Patrick Baumgartn...	11/6/23, 5:49:09 PM	N/A	—	Confidential\Micro... ⓘ	
	E-Bike Rental Analysis	Notebook	Patrick Baumgartn...	—	—	—	Confidential\Micro... ⓘ	



Generative AI on your data

Announcing Public Preview

Deliver custom
generative AI experiences
for **your data**



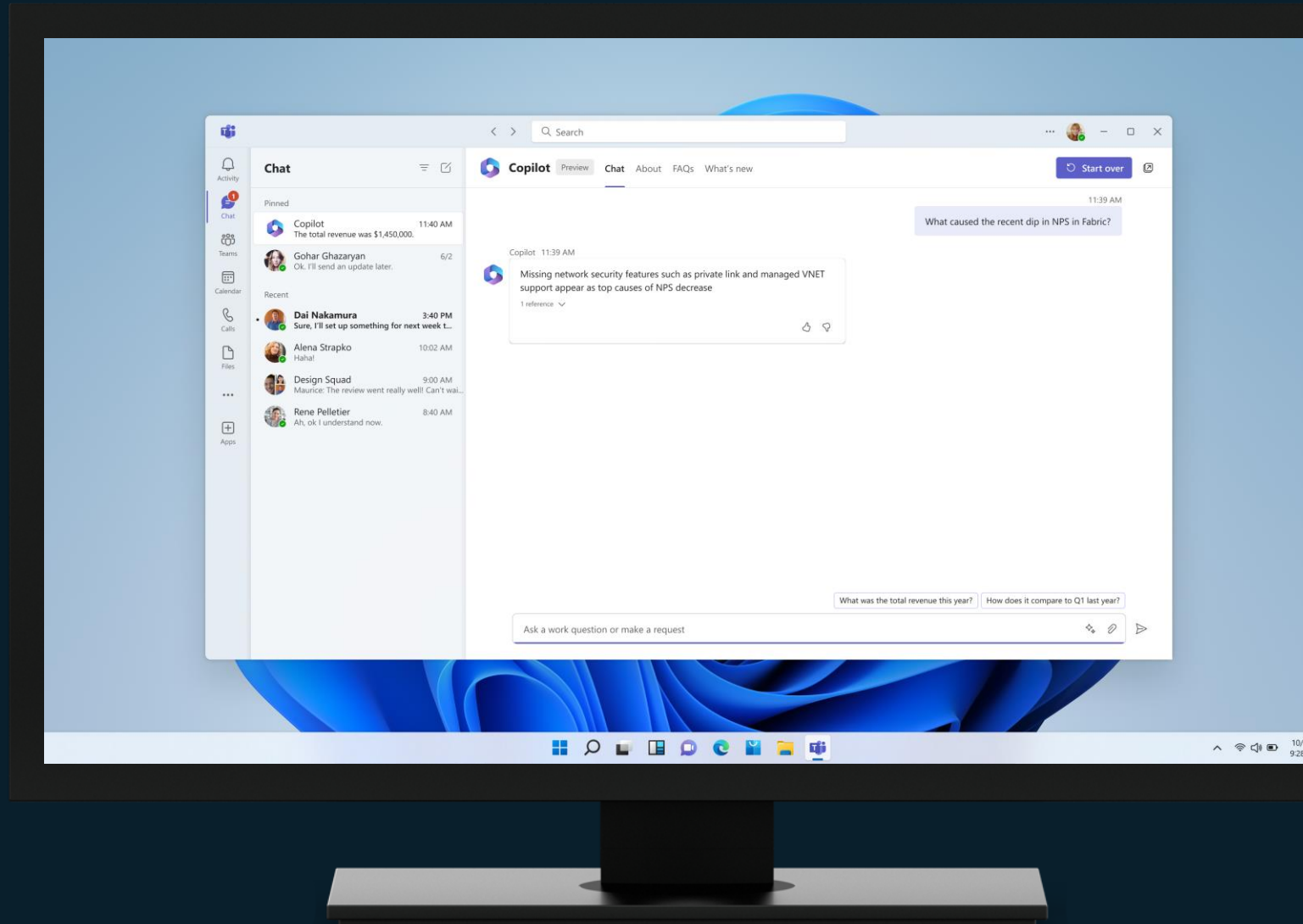
Enable custom Q&A on your data in Fabric



Define custom business semantics and grounding **unique to your org**



Scale the custom experiences to **M365 Chat, Copilot Studio, and Azure AI Studio**



Next steps



Try Microsoft
Fabric

aka.ms/try-Fabric



Register for the
**Microsoft
Fabric Community
Conference**

aka.ms/FabricConference

Further resources



Explore the product here:

<https://www.microsoft.com/en-us/microsoft-fabric>



Check out the mechanics show:

<https://youtu.be/oxMUJT3p9f4>



Explore the documentation:

<https://learn.microsoft.com/en-us/fabric/>



Read the free eBook: Getting started with Fabric

<https://aka.ms/fabric-get-started-ebook>

End-to-end tutorials

Lakehouse tutorial

<https://learn.microsoft.com/en-us/fabric/data-engineering/tutorial-lakehouse-introduction>

Data Science tutorial

<https://learn.microsoft.com/en-us/fabric/data-science/tutorial-data-science-introduction>

Real-Time Analytics tutorial

<https://learn.microsoft.com/en-us/fabric/real-time-analytics/tutorial-introduction>

Data warehouse tutorial

<https://learn.microsoft.com/en-us/fabric/data-warehouse/tutorial-introduction>

Power BI tutorial

<https://learn.microsoft.com/en-us/power-bi/fundamentals/fabric-get-started>

Data Factory tutorial

<https://learn.microsoft.com/en-us/fabric/data-factory/tutorial-end-to-end-introduction>



Thank you



Appendix